

CPR Esoft Quiz-6

Question #: 1

A 60-year-old woman comes to the physician because of a 2-month history of progressive weakness, loss of appetite and nausea. Her temperature is 36.5°C (97.7°C), pulse is 84/min, respirations are 20/min, and blood pressure is 160/90 mm Hg. Physical examination shows marked paleness and pitting pedal edema. Her 24-hour urine volume is 576 mL, urine creatinine concentration is 230 mg/dL, and plasma creatinine is 8 mg/dL. Which of the following values best approximates her creatinine clearance?

- A. 3 mL/min
 - B. 6 mL/min
 - C. 9.5 mL/min
 - D. 11.5 mL/min
 - E. 15 mL/min
-

Question #: 2

A 50-year-old man comes to the physician because of a 1-week history of increased thirst and frequent urination. He was diagnosed with diabetes insipidus 1 year ago. His vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following free water clearance values is most likely in this patient?

- A. -3
 - B. -2
 - C. -1
 - D. 0
 - E. +2
-

Question #: 3

A 38-year-old man comes to the physician because of a 2-week history of increased thirst and frequent urination. He has no major illness. His vital signs are all within normal limits. Physical examination shows no abnormalities. Laboratory studies show a hypotonic urine and decreased serum antidiuretic hormone concentration. A water deprivation test is performed, and an antidiuretic hormone is administered. Following this administration, his urine osmolality markedly increased. Which of the following is the most likely diagnosis?

- A. Central diabetes insipidus

- B. Nephrogenic diabetes insipidus
 - C. Type 1 diabetes mellitus
 - D. Syndrome of inappropriate diuretic hormone (SIADH)
 - E. Primary polydipsia
-

Question #: 4

A 31-year-old woman comes to the physician because of a 3-week history of increased thirst and urination. She says she just recovered from a respiratory infection that was treated with a 10-day course of antibiotics. She has no major illness. Her blood pressure is 114/74 mm Hg; other vital signs are within normal limits. Physical examination shows dry tongue and mouth, and poor skin turgor. Laboratory studies show increased serum concentration of sodium and osmolality, reduced urine osmolality, and absent glucose in urine. She is diagnosed with nephrogenic diabetes insipidus because of antibiotic treatment. Which of the following renal changes is most likely in this patient?

- A. Increased GFR
 - B. Decreased serum concentration of ADH
 - C. Decreased urea concentration in the medullary interstitium
 - D. Diminished reabsorption of Na⁺ in PCT
 - E. Reduced reabsorption of glucose in PCT
-

Question #: 5

A 23-year-old man comes to the physician because of a 2-day history of passing large volume of urine and increased thirst. Three weeks ago, he was involved in a road traffic accident and hospitalized for a week because of head injury. His vital signs are within normal limits. Laboratory studies show a low serum concentration of ADH. Which of the following renal alterations is most likely because of the ADH deficiency in this patient?

- A. Increased glomerular filtration rate
 - B. High osmolarity of tubular fluid leaving the thick ascending limb
 - C. Reduced urea reabsorption in the proximal tubule
 - D. Elevated renal medullary osmotic gradient
 - E. Low osmolarity of tubular fluid at the end of the cortical collecting duct
-

Question #: 6

A 63-year-old woman is brought to the physician by her husband because of a 2-day history of forgetfulness and confusion. She has no other medical issues and is not on any regular medications. Her husband says that she has been drinking a lot of plain water for the past few days. Her blood pressure is 120/66 mm Hg, and pulse is 82/min. She is mildly disoriented and

confused. Laboratory tests show that her blood glucose is normal, but serum sodium concentration is 128 mEq/L (Normal: 135-145 mEq/L). Which of the following best explains this patient's altered mental status?

- A. Decreased intracellular fluid volume
 - B. Increased intracellular osmolarity
 - C. Decreased plasma osmolarity
 - D. Increased plasma osmolarity
 - E. Low blood pressure
-

Question #: 7

A 35-year-old man is brought to the emergency department following a car crash that occurred an hour ago. His temperature is 35.1°C (95.2°F), pulse is 110/min, blood pressure is 90/60 mm Hg, and respirations are 22/min. Physical examination shows open fractures of his left radius and ulna. Laboratory studies show a hematocrit of 32% and a plasma volume of 3 Liters. Which of the following values best approximates the blood volume of this patient?

- A. 3.3 L
 - B. 4.0 L
 - C. 4.3 L
 - D. 4.4 L
 - E. 5.4 L
-

Question #: 8

A 31-year-old man participates in a study related to estimation of various body fluids. His vital signs are within normal limits. He is injected intravenously with a radioactive sodium solution. The data during the study are tabulated as below:

Amount of radioactive sodium injected = 380 mmol

Amount of radioactive sodium lost in urine within an hour = 20 mmol

Concentration of radioactive sodium in serum after one hour = 0.02 mmol/L.

Assuming that the radioactive sodium is distributed in his body fluid during 1-hour equilibration time. Which of the following values best represents the extracellular fluid volume in this person?

- A. 15 Liter
 - B. 18 Liter
 - C. 24 Liter
 - D. 30 Liter
 - E. 42 Liter
-

Question #: 9

A 28-year-old man comes to the physician for a routine examination 1 month after surgery to remove a kidney stone. His blood pressure is 120/76 mmHg. The physician receives the following laboratory data from his recent renal clearance study:

Hydrostatic pressure in glomerular capillaries = 59 mm Hg

Hydrostatic pressure in Bowman's Space = 18 mm Hg

Oncotic pressure of filtrate in Bowman's Space = 1 mm Hg

Oncotic pressure of plasma in glomerular capillaries = 32 mm Hg

Filtration coefficient (K_f) = 11 mL/min/mm Hg

RPF = 600 mL/min

Which of the following values best approximates this patient's filtration fraction?

- A. 0.10
 - B. 0.18
 - C. 0.2
 - D. 0.25
 - E. 0.5
-

Question #: 10

A 16-year-old girl is brought to the emergency department by her parents because of a 3-day history of fever, shortness of breath, muscle pain, and difficulty walking. Her blood pressure is 160/120 mm Hg, and pulse is 84/min. An oral administration of an anti-hypertensive drug and a beta-blocker is not effective in reducing her blood pressure. Ultrasound imaging of kidneys shows decrease in the size of the kidneys and calcification of the renal arteries, causing bilateral renal artery stenosis. Urinalysis shows significantly reduced urine output because of a reduction in the glomerular filtration. A decrease in which of the following best explains the reduced GFR in this patient?

- A. Afferent arteriolar resistance
 - B. Glomerular capillary filtration coefficient
 - C. Protein concentration in Bowman's space
 - D. Efferent arteriolar resistance
 - E. Glomerular capillary hydrostatic pressure
-

Question #: 11

A 25-year-old man is brought to the emergency department by a rescue team from a mountain expedition site. His blood pressure is 80/62 mm Hg, and pulse is 86/min. He is conscious but his lips, mouth and eyes are dried because of severe dehydration. His urine output has decreased significantly. There is no proteins lost in his urine. Serum studies show elevated angiotensin II concentration. This hormone causes vasoconstriction of the efferent arterioles. Which of the following changes of the Starling forces at the peritubular capillary is most likely in this patient?

- A. Hydrostatic pressure increases
 - B. Hydrostatic pressure does not change
 - C. Oncotic pressure does not change
 - D. Oncotic pressure decreases
 - E. Oncotic pressure increases
-

Question #: 12

A 25-year-old man is brought to the emergency department 1 hour after he was involved in a motor vehicle collision in which he sustained burn injuries. His temperature is 37°C (98.6°F), blood pressure is 98/62 mmHg, pulse is 124/min, and respirations are 20/min. Physical examination shows second, and third degree burns on his face, trunk, and arms. He is diagnosed with hypovolemic shock and is resuscitated with intravenous fluids. Four hours later, his urine output significantly decreases, and his blood pressure does not improve. Serum studies show electrolytes and albumin concentrations within normal range. Which of the following best explains a reduction in the urine output of this patient?

- A. Increased glomerular capillary hydrostatic pressure
 - B. Decreased glomerular capillary hydrostatic pressure
 - C. Increased glomerular capillary oncotic pressure
 - D. Decreased glomerular capillary oncotic pressure
 - E. Increased glomerular filtration coefficient
-

Question #: 13

A 30-year-old man is brought to the emergency department by his wife because of a 2-hour history of chest pain and shortness of breath. An ECG shows flat T waves in multiple leads. Laboratory studies show a serum potassium concentration of 3 mmol/L (normal range 3.5 - 5 mmol/L). Imaging his kidney and urinary bladder shows cortical damage of the right kidney, leading to the poor reabsorption of potassium from the kidney tubules. No pathology is visible in the left kidney. Which of the following mechanisms best explains potassium reabsorption in the proximal tubules of this patient's left (normal) kidney?

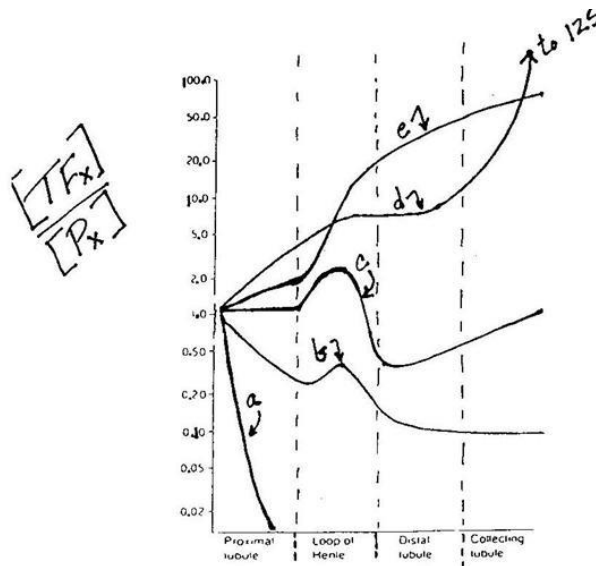
- A. Occurs predominantly by solvent drag via paracellular pathways
 - B. Is coupled to H⁺ secretion
 - C. Accounts for less than 50% of K⁺ reabsorption in the nephron
 - D. Is sensitive to circulating levels of ADH
 - E. Is coupled to the reabsorption of glucose and amino acids
-

Question #: 14

A 32-year-old woman comes to the physician because of a 1-week history of back pain and painful urination. She has a history of frequent urinary tract infections. An ultrasound of the kidneys and urinary bladder indicates cortical scars. Her tubular fluid/plasma analysis graph is attached. Which of the following best represents the relative concentration of urea along her kidney tubules?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



Renal Analysis_17.jpg

Question #: 15

A 46-year-old man comes to the physician because of a 1-month history of fatigue and shortness of breath. His blood pressure is 160/110 mm Hg. Physical examination shows generalized pitting edema. Results of laboratory studies of his blood and urine are shown:

Plasma creatinine: 1.5 mg/dL (normal: 0.6-1.2 mg/dL)

Glycosylated hemoglobin (HbA1c): 5.8% (normal: <6%)

Urine creatinine: 200 mg/dL or 2 mg/mL

24-hour urine volume: 1.44 L or 1 mL/min

Urine PAH: 250 mg/dL

Which of the following values best approximates his GFR?

- A. 80 mL/min
- B. 98 mL/min
- C. 106 mL/min

- D. 126 mL/min
 - E. 133 mL/min
-

Question #: 16

A 62-year-old man comes to the physician because of a 3-day history of headache, mild chest pain on short walks, difficulty breathing and occasional blurred vision. His blood pressure is 164/96 mmHg, and pulse is 68/min. His plasma electrolytes are within the normal ranges. He is prescribed a diuretic drug that inhibits sodium-chloride reabsorption in the distal convoluted tubules. Which of the following groups of diuretics is most likely prescribed to this patient?

- A. Osmotic diuretic
 - B. Loop diuretic
 - C. Potassium sparing diuretic
 - D. Thiazide diuretic
 - E. Carbonic anhydrase inhibitor
-

Question #: 17

A 25-year-old man comes to the physician because of a 10-day history of breath odor and headaches. He says that he adopted a ketogenic diet for weight loss 4 weeks ago. He is 180 cm tall and weighs 130 kg; BMI is 40.1 kg/m². Physical examination shows a fruity odor to his breath. Laboratory studies show serum glucose concentration is 75 mg/dL (Normal: 70-100 mg/dL). Urinalysis is positive for ketone bodies. Which of the following best describes his acid-base balance on this diet?

- A. The renal carbonic anhydrase system is less active
 - B. Secreted protons are buffered by NH₃ to form NH₄Cl in the collecting duct
 - C. Urine pH is increased, and HCO₃⁻ is lost in urine
 - D. Renal glutaminase activity is very low
 - E. Beta-intercalated cells secrete H⁺ ions into tubular lumen and reabsorb bicarbonate
-

Question #: 18

An 85-year-old woman is brought to the physician by her daughter because of a 3-day history of chest discomfort and generalized malaise. Two weeks ago, she developed pedal edema, and her physician prescribed a loop diuretic. She has a history of hypertension, type 2 diabetes mellitus, dyslipidemia, and Parkinson's disease and reports compliance with her medications. Which of the following acid-base disturbances is most likely to be observed in this patient after initiation of the diuretic?

- A. Metabolic acidosis with hyperkalemia
 - B. Increased anion gap metabolic acidosis
 - C. Metabolic alkalosis with hypokalemia
 - D. Respiratory alkalosis with renal compensation
 - E. Acute respiratory acidosis
 - F. Respiratory acidosis with renal compensation
-

Question #: 19

An 18-month-old boy is brought to the physician by his parents because of a 3-day history of severe cough, shortness of breath, and fever. His medical history is notable for recurrent respiratory infections and chronic diarrhea since 6 months of age, requiring multiple hospital admissions. Laboratory studies show very low B-cell counts. A serum protein electrophoresis is performed, confirming a primary immunodeficiency. Which of the following protein fractions is most likely decreased in this patient?

- A. Albumin
 - B. Alpha-1 globulin
 - C. Alpha-2 globulin
 - D. Beta globulin
 - E. Gamma globulin
-

Question #: 20

A 58-year-old man has been admitted to the hospital for evaluation of jaundice and abdominal swelling for the past month. He has a 15-year history of alcohol use disorder. Physical examination shows jaundice and ascites. Laboratory studies show an AST: ALT ratio of 3:1. Serum protein electrophoresis is performed. Which of the following abnormalities in the serum protein profile is most likely observed in this patient?

- A. A globulin beta-gamma bridge
 - B. A highly elevated albumin peak
 - C. A single elevated peak of monoclonal immunoglobulin
 - D. An elevated alpha-1 antitrypsin peak
 - E. A sharply elevated peak of alpha2-globulins
-

Question #: 21

A 26-year-old man is admitted to the hospital for evaluation of a 2-week history of fever, chills, and generalized weakness. Physical examination shows pallor. Laboratory studies show hemoglobinuria. Peripheral blood smear shows intraerythrocytic ring forms consistent with Plasmodium species. To assess the severity of hemolysis, the concentration of a specific serum

protein is measured. Which of the following best describes this serum protein?

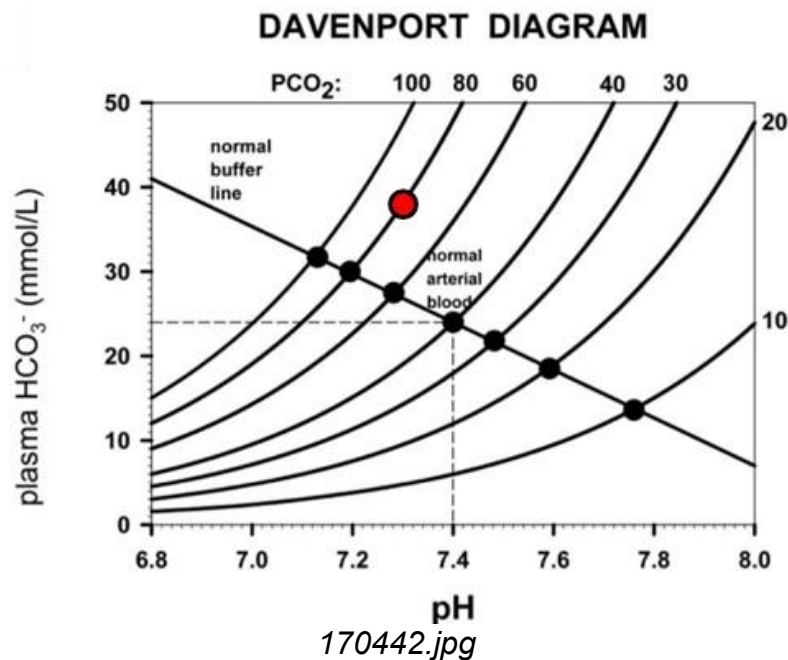
- A. Binds to free heme in plasma
- B. Binds to free hemoglobin in plasma
- C. Transports iron in plasma
- D. Negative acute-phase reactant
- E. Protein of the beta-globulin fraction

Question #: 22

A 58-year-old man is brought to the physician because of a 2-day history of worsening shortness of breath. His wife says that he has had a fever and a productive cough for 5 days. He has a 25-pack-year history of cigarette smoking. Physical examination shows cyanosis and respiratory distress. Auscultation of the lungs shows diffuse crepitations bilaterally. Arterial blood gas studies on room air are represented by the red dot on the Davenport diagram shown. Which of the following compensatory mechanisms is most likely active in this patient?

- A. Decreased serum bicarbonate levels
- B. Increased urinary bicarbonate excretion
- C. Inhibition of renal carbonic anhydrase
- D. Increased urinary ammonium excretion
- E. Increased rate and depth of respiration

Attachment:



Question #: 23

A 45-year-old man presents to the emergency department with a 2-week history of fatigue and muscle cramps. Laboratory studies show metabolic alkalosis, increased renal tubular H^+ ion secretion, and a persistently low urine pH. Which of the following conditions is most likely responsible for these findings?

- A. Decreased arterial PCO_2
 - B. Adrenal cortical insufficiency
 - C. Administration of carbonic anhydrase inhibitor
 - D. Diabetic ketoacidosis
 - E. Hypokalemia
-

Question #: 24

A 4-year-old boy is brought to the emergency department by his mother because of a 1-hour history of abdominal pain and vomiting. She says that she found an empty bottle of aspirin tablets (acetyl salicylic acid). His respirations are 32/min. Pulse oximetry on room air shows oxygen saturation of 95%. Arterial blood gas analysis on admission show:

- pH = 7.5 (Normal: 7.35-7.45)
- PCO_2 = 30 mm Hg (Normal: 33-45 mm Hg)
- Bicarbonate (HCO_3^-) = 23 mEq/L (Normal: 22-28 mEq/L)

Which of the following acid-base disorders is most likely in this patient?

- A. Chronic respiratory acidosis due to poor tissue oxygenation
 - B. Acute respiratory alkalosis due to hyperventilation
 - C. Metabolic acidosis due to loss of bicarbonate
 - D. Chronic respiratory alkalosis with renal compensation
 - E. Acute respiratory acidosis due to hypoventilation
 - F. Mixed alkalosis
-

Question #: 25

A 20-year-old man is brought to the emergency department after a motor vehicle collision. His blood pressure is 70/60 mm Hg, and pulse is 100/min. Physical examination shows a rapid and weak pulse, multiple long-bone fractures, and extensive bruising of the extremities. Initial laboratory and arterial blood gas studies on room air are shown in the table. Which of the following is the most likely cause of this acid-base disturbance?

- A. Lactic acidosis
- B. Respiratory distress syndrome
- C. Nasogastric suction

- D. High altitude sickness
- E. Pyloric stenosis
- F. Diarrhea

Attachment:

Analyte	Patient	Reference range
Na ⁺	142	136-145 mEq/L
Cl ⁻	98	95-105 mEq/L
PaCO ₂	25	33-45 mmHg
HCO ₃ ⁻	10	22-28 mmol/L
pH	7.22	7.35 – 7.45

170458.jpg

Question #: 26

A medical student reviews arterial blood gas studies for a patient:

- pH: 7.50 (normal: 7.35–7.45)
- PaCO₂: 48 mm Hg (normal: 33-45 mm Hg)
- Bicarbonate (HCO₃⁻): 32 mmol/L (normal: 22–28 mmol/L)

Which of the following clinical conditions best explains these findings?

- A. A 3-week-old boy with a 4-day history of projectile vomiting
- B. A 45-year-old man who has been on treatment for chronic obstructive lung disease presenting with fever, cough, and chest pain
- C. A 20-year-old woman who has been camping at a high altitude for the past two days and has been hyperventilating
- D. A 20-year-old man with type 1 diabetes mellitus with ketoacidosis
- E. A 60-year-old man who has an airway obstruction due to a tumor in the right bronchus
- F. A 2-year-old boy with severe diarrhea for three days

Question #: 27

A 50-year-old man undergoes elective surgery and receives preanesthetic medication containing morphine. Two hours later, he develops pinpoint pupils, shallow respirations, and confusion. Respirations are 6/min. Physical examination shows slow and shallow breathing. Which of the following sets of arterial blood gas (ABG) findings is most likely observed in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V

Attachment:

	pH	PCO ₂	HCO ₃ ⁻
I	Normal	Increased	Decreased
II	Increased	Decreased	Increased
III	Decreased	Increased	Normal
IV	Increased	Increased	Increased
V	Decreased	Decreased	Normal

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Question #: 28

A 16-month-old boy is brought to the physician by his mother because of a 3-day history of fever and severe diarrhea. His pulse is rapid and weak, blood pressure is low, and respiratory rate is increased. Physical examination shows sunken eyes and decreased skin turgor. Arterial blood gas studies on room air are obtained. Which of the following sets of findings is most likely to be observed in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V
- F. VI

Attachment:

	pH	PCO ₂	HCO ₃ ⁻	Anion gap
I	Increased	Decreased	Decreased	Normal
II	Decreased	Decreased	Decreased	Normal
III	Decreased	Increased	Increased	Increased
IV	Decreased	Decreased	Decreased	Increased
V	Decreased	Normal	Decreased	Normal
VI	Increased	Increased	Increased	Increased

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Question #: 29

A 27-year-old man comes to the physician because of a 2-day history of headache and nausea. He recently moved from New Jersey (sea level) to Colorado Springs, which is at 4300 m (14,000 ft) above sea level. Arterial blood gas studies on room air show:

- pH: 7.49 (Normal: 7.35–7.45)
- PCO₂: 25 mm Hg (Normal: 33–45 mm Hg)
- Bicarbonate (HCO₃⁻): 24 mmol/L (Normal: 22–28 mmol/L)

Which of the following changes is most likely observed after one week in this patient?

- A. An increase in urinary bicarbonate
- B. Inhibition of erythropoietin secretion
- C. Stimulation of renal carbonic anhydrase
- D. An increase in urinary acid phosphate excretion
- E. An increase in renal tubular hydrogen ion secretion
- F. Decreased erythrocyte 2,3-bisphosphoglycerate levels

Question #: 30

A 13-year-old girl is brought to the emergency department by her parents because of a 1-day history of worsening nausea and abdominal pain. She has a 3-week history of increased thirst and urination. Laboratory studies on admission show:

- Serum glucose: 550 mg/dL (Normal: 70-100 mg/dL)

- Urinalysis: positive for glucose and ketone bodies

Which of the following arterial blood gas (ABG) results is most likely to be observed in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V
- F. VI

Attachment:

	pH	PCO ₂	HCO ₃ ⁻	Anion Gap
I	Decreased	Normal	Decreased	Increased
II	Decreased	Decreased	Increased	Normal
III	Decreased	Decreased	Decreased	Increased
IV	Decreased	Decreased	Decreased	Normal
V	Decreased	Increased	Normal	Increased
VI	Increased	Increased	Increased	Normal

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